



SUBJECT	Lower Secondary Science Year 8
TOPIC	Energy
ASSIGNMENT	Six
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: _____

ASSIGNMENT GUIDANCE:

- You should complete this assignment by carrying out Practical 3: Insulation on this sheet and then handwriting in black ink both the Practical 3: Insulation questions and the assignment questions that follow on, unless you have obtained specific permission to submit typed work. Then scan it in and submit it, as a single document.
- This is not a timed assignment. We estimate that this assignment may take you between 1 and 2 hours for the practical, and between 40 and 60 minutes for the assignment itself. However, if you need to take longer, then please do so.
- Make sure you write your full name clearly in the box at the top of this page.
- In order to complete this assignment, you will need a calculator, ruler, your coursebook and notes.





LS Science Assignment Six

PRACTICAL 3: INSULATION

1. How does the type of insulation affect the rate of heat loss?

Different materials will be better at keeping heat in than others. You can do a simple experiment to test which of your clothes, for example wool versus manmade fibre, would be the most insulating, by measuring the rate of heat loss from a cup of hot water wrapped in a layer of the material.

Apparatus

You will need:

- Two different types of material. ask a responsible person for pieces of material that are not being used, so that you can cut them to size - things like jumpers with holes in them, or old towels would be best, but please ask permission before you use them! Try to choose very different types of material to get the biggest difference in results. If you have no cloth material, you could use newspaper and aluminium foil instead.
- If possible, make the material about 1cm thick around the container (this will not be possible with newspaper or aluminium foil, but you could put 5 layers on to make it more effective)
- Three identical cups. If this is not possible, just use one cup and repeat the experiment three separate times.
- Some thick cardboard to make lids with a small hole for a thermometer (you could use disposable coffee cups and plastic lids)
- Thermometer
- Elastic bands to hold the material onto the cup.
- Kettle
- Stopwatch
- Permanent marker. if you are allowed to draw on the cup with it.

Method

Read the whole thing through before you start and answer the questions as you go. You will need to design your own table for your results (see Question below) before you get started.

- Cut your pieces of material to the correct size, so you get exactly one layer wrapped around your cup (a little overlap is fine, just try to get the pieces exactly the same size).
- Add a line to the inside of the cup with the permanent marker so that you always use the same volume of water.
- Add boiling water to the cup, fit the lid on top, and use the thermometer to record the temperature at time zero. Start the stopwatch





- Record the temperature every minute for ten minutes. Stir the water before you take your reading so that the temperature is the same everywhere in the water (making the results reliable).
- Repeat with the other piece of material.
- Repeat with **no material** around the cup.

Variables

- a. What are the dependent and independent variables in this experiment?

(2 marks)

- b. Why do you have to keep the material the same size and the volume of water the same?

(1 mark)

- c. What is this repeat of the experiment called, with nothing around the cup?

(1 mark)





Results

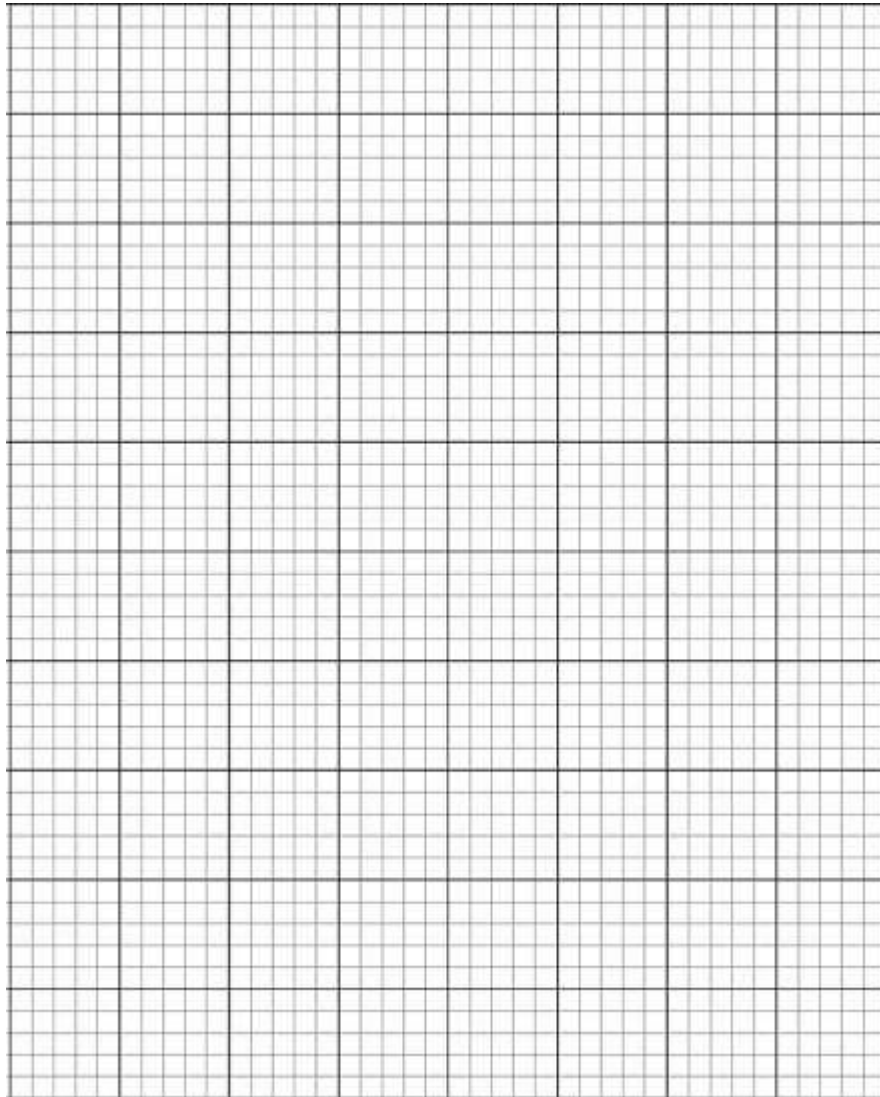
- d. Use this space to draw a table in which to record your results. Look back in the method to remind yourself of the variables, and how many readings you will need to take.

(3 marks)



e. Now draw a graph of your results.

- Put time on the horizontal axis and the dependent variable on the vertical axis. Your independent variables will each be represented by different curves, so you might want to use different colours to draw them.
- Do not forget to use a regular scale on both axes. Make sure you use a scale that stretches out so you use as much of the graph space as you can.
- Plot your points carefully and draw a line of best fit for each independent variable. It might be a good idea to draw your line of best fit in pencil, and when you are satisfied with one, draw over it in pen before moving on to the next one. That would be safer than drawing them in pen straight away!



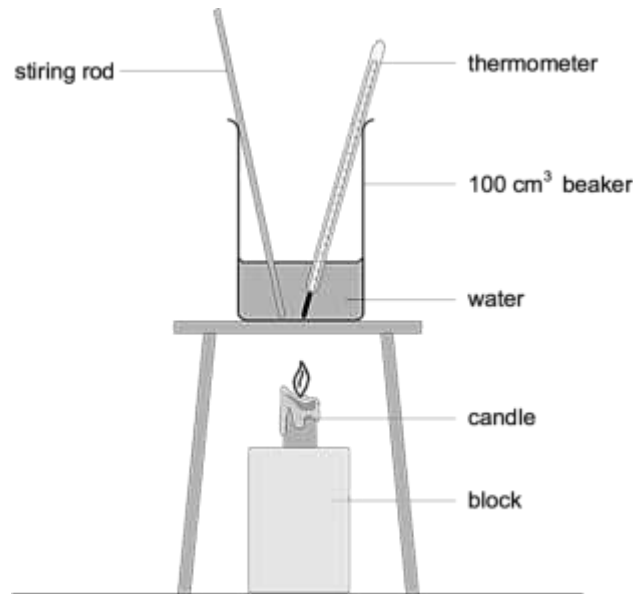
(5 marks)

(Total 12 marks)

ENERGY

2. Luke investigated the heating of water. He predicted that the rise in temperature would depend on the volume of water.

The diagram shows the apparatus he used.



Luke recorded his results in a table as shown below.

beaker	volume of water, in cm ³	temperature at start, in °C	temperature after 2 minutes, in °C
A	25	18	30
B	50	18	24
C	75	18	22

- a. Why did Luke need to know the temperature of the water at the beginning and at the end of the experiment?

(1 mark)

- b. Did Luke's results support his prediction? Explain your answer.

(1 mark)



- c. Luke stirred the water during the experiment. How did this make his results more reliable?

(1 mark)

- d. Which of the following statements about the energy transferred to the beakers is correct?
Tick the correct box.

Much more energy went into beaker 'A'
because its temperature increased the most.

☐

The same amount of energy went into all
three beakers.

☐

Beaker 'C' received the most energy
because there was more water to heat.

☐

(1 mark)

- e. After a time, all three beakers cooled down to room temperature.
What happened to the thermal energy in the beakers as they cooled down?

(1 mark)

- f. Luke decided to repeat his experiment and produced 2 more sets of results. What should he
do with these results? Does this make his results more reliable?

(2 marks)

(Total 7 marks)



3. The drawings below show what happens to the energy supplied to four appliances.

 kettle	98% of the energy is used to heat the water 2% of the energy is wasted
 light bulb	% of the energy is given out as light 95% of the energy is wasted
 radio	50% of the energy is given out as sound 50% of the energy is wasted
 mixer	40% of the energy is used to move the whisks % of the energy is wasted

a. What percentage of energy of the light bulb is given out as light?
Write your answer on the line by the light bulb. (1 mark)

b. What percentage of energy is wasted by the mixer?
Write your answer in the box by the mixer. (1 mark)

c. Complete the sentence below.

Parts of the mixer become hot because some of the electrical energy is changed into _____ energy which is wasted.

(1 mark)

- a. Energy is wasted as sound in many appliances. Which appliance in the drawings produces sound which is **not** wasted?

(1 mark)

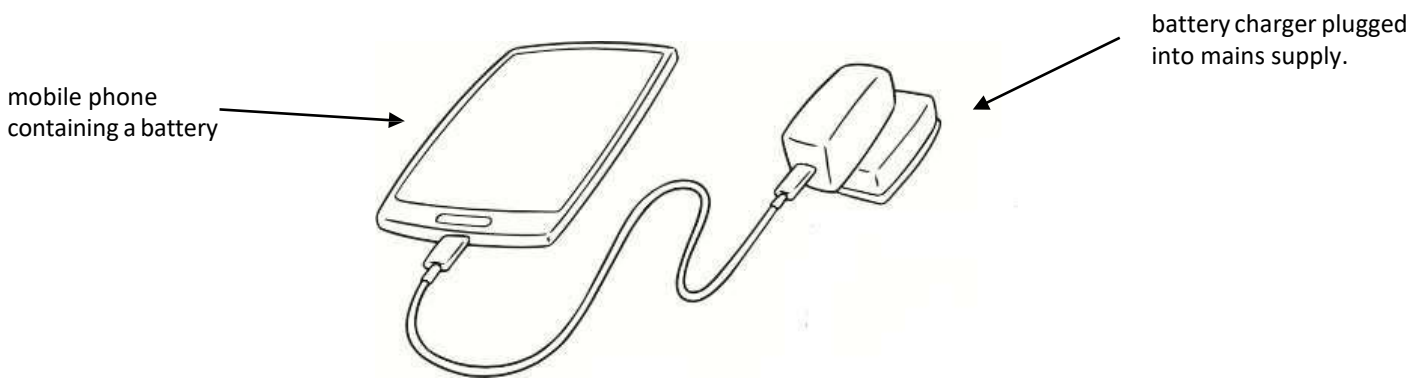
- b. In which of the appliances is the highest percentage of energy wasted?

(1 mark)

(Total 5 marks)

4.

- a. Jacquie has a mobile phone. Energy is stored in the battery of the phone. The drawing shows the battery being charged.



Which energy transfer takes place in the battery as it is being charged? Tick the correct box.

chemical to sound	☐	sound to thermal	☐
electrical to chemical	☐	thermal to electrical	☐

(1 mark)

- b. When the battery is fully charged, Jacquie unplugs the phone. Which energy transfers take place when the mobile phone rings? Tick the correct box.



chemical to electrical to sound	☐
electrical to chemical to sound	☐
kinetic to electrical to sound	☐
thermal to electrical to sound	☐

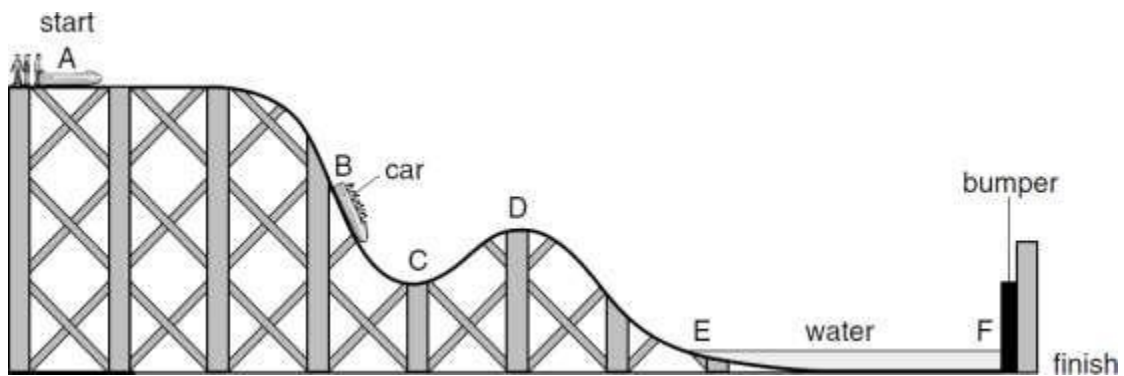
(1 mark)

(Total 2 marks)

5. The photograph shows some pupils in a log car on a theme-park ride.



The drawing below shows the ride. The letters A, B, C, D, E and F show different points along the track.



The car starts from A and travels to F, where it stops by hitting a bumper. At E the car enters a trench filled with water.

a.

- i. At which **two** points does the car have **no** kinetic energy? Give the **two** correct letters.

_____ and _____

(1 mark)

- ii. At which point does the car have the **most** gravitational potential energy? Give the correct letter.

(1 mark)



- iii. At which point does the car have **some** kinetic energy and the **least** gravitational potential energy? Give the correct letter

(1 mark)

b.

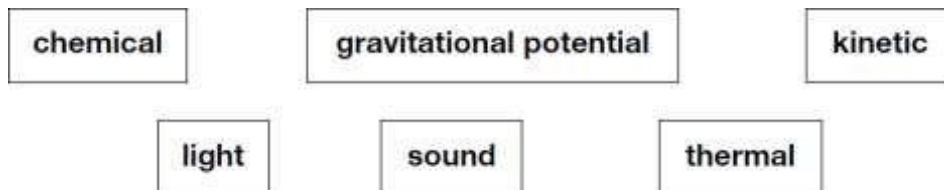
- i. The cars are **not** powered by a motor. What force causes the cars to move along the track from B to C?

(1 mark)

- ii. When a car splashes through the water at E, it slows down. What force acts on the car to slow it down?

(1 mark)

- c. Complete the sentence below by choosing from the following words.

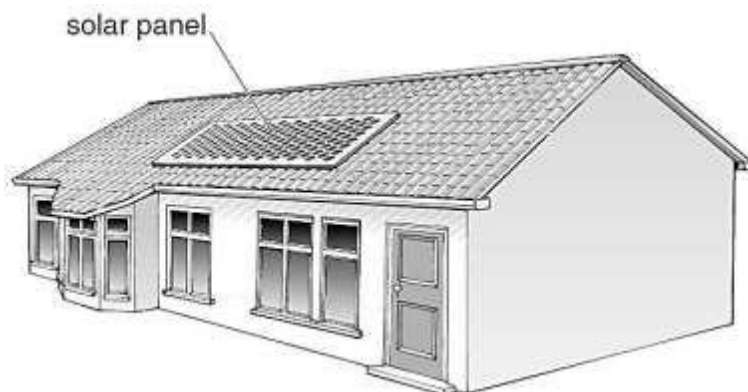


When the car hits the bumper at F, its _____ energy is transferred into _____ energy and _____ energy.

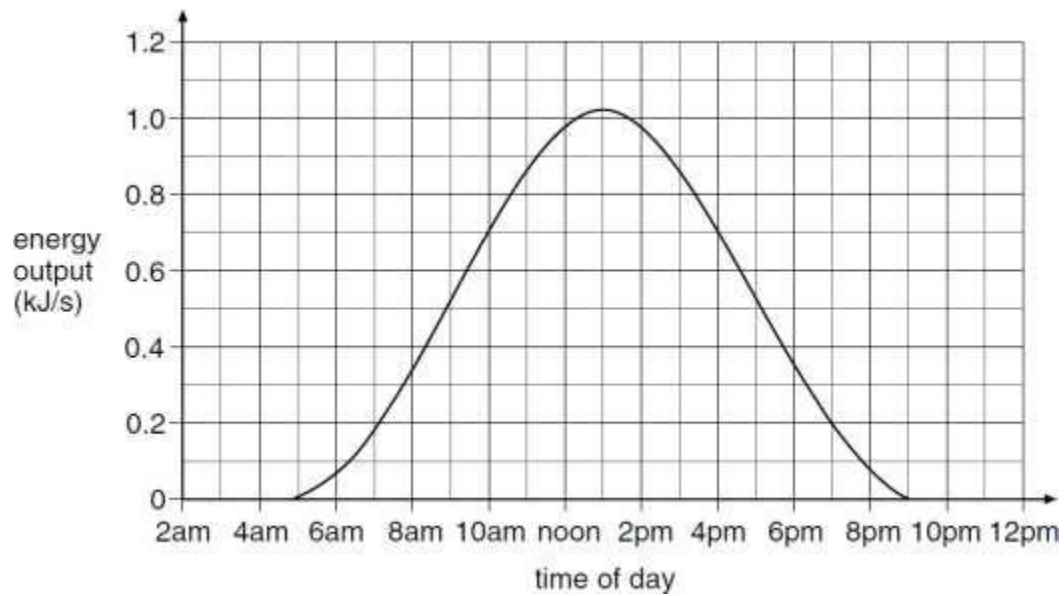
(3 marks)

(Total 8 marks)

6. The drawing below shows a solar panel fixed to the roof of a house in Britain.



- a. Daniel measured the energy output from the solar panel during one day in June. The graph below shows his results.



- i. Why does the energy output from the solar panel vary during the day?

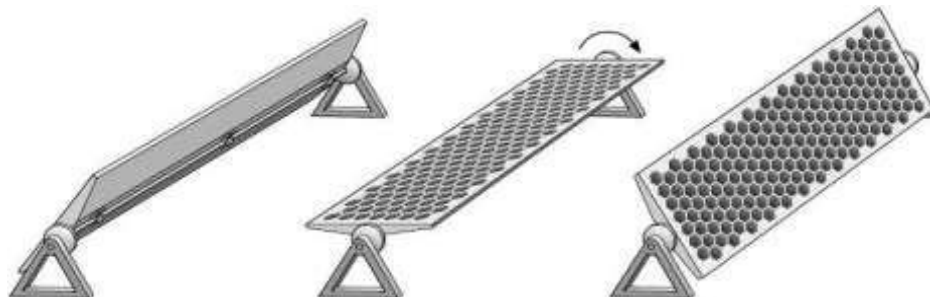
(1 mark)

- ii. Daniel used the solar panel to run a motor. The motor needs 0.7kJ/s to run at full speed. Use the graph to find how long Daniel's motor would run at full speed.

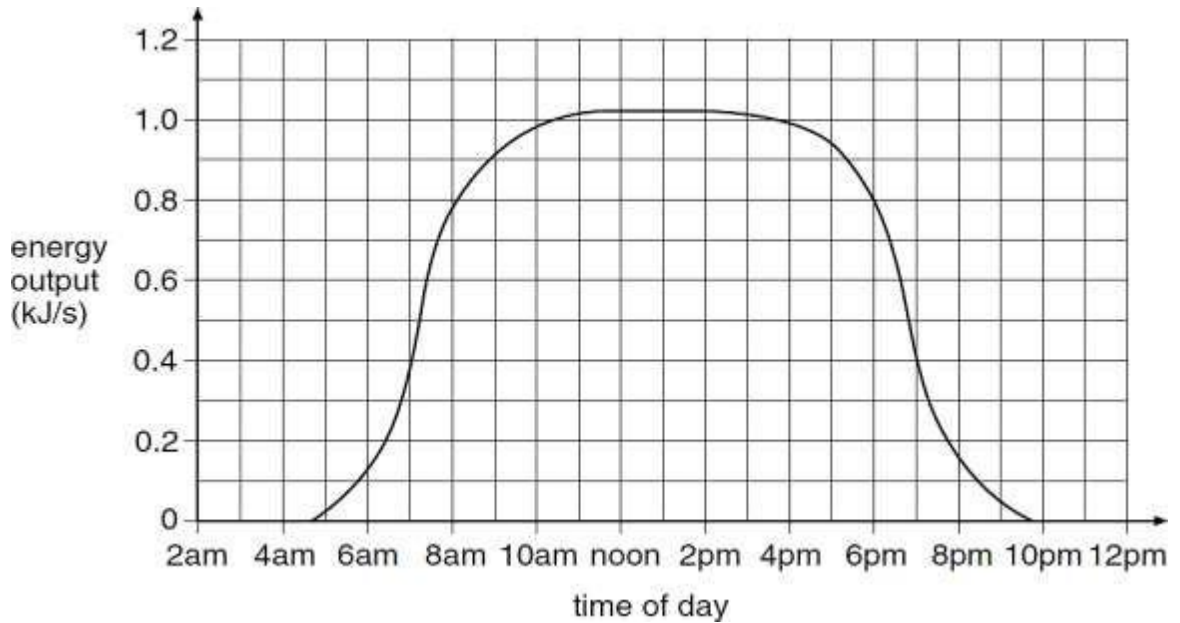
_____ hours

(1 mark)

- b. Daniel measured the energy output from a different solar panel. This type of solar panel turns so that it always faces the sun.



The graph below shows the energy output for this panel during one day in **mid-summer**.



- i. On the graph above draw another curve to show how the energy output for this solar panel might vary on a day in **mid-winter**.

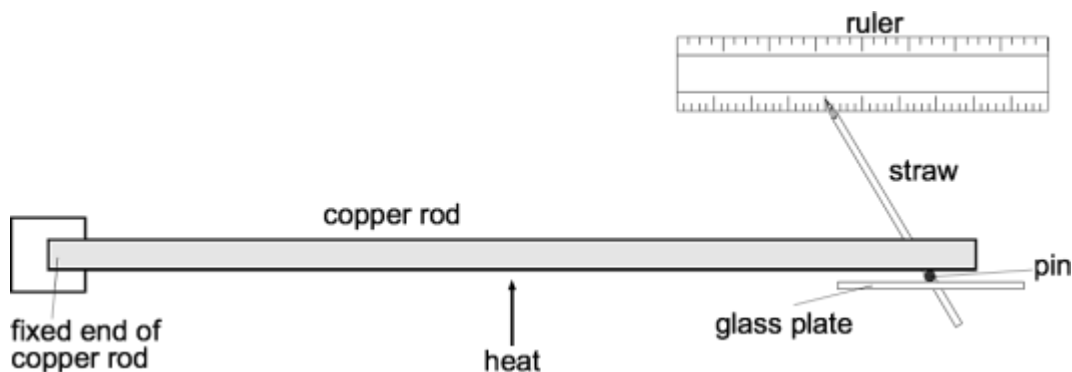
(2 marks)

- ii. Between 7am and 7pm the solar panel turns through an angle of 180° . Calculate the angle (in degrees) that the solar panel turns through each hour.

(1 mark)

(Total 5 marks)

7. Sarah wanted to find out how the length of a copper rod changes when it is heated. She fixed one end of the rod and allowed the other end to roll on a pin which had been put on a glass plate. A drinking straw, attached to the pin, showed the movement of the end of the rod.



a.

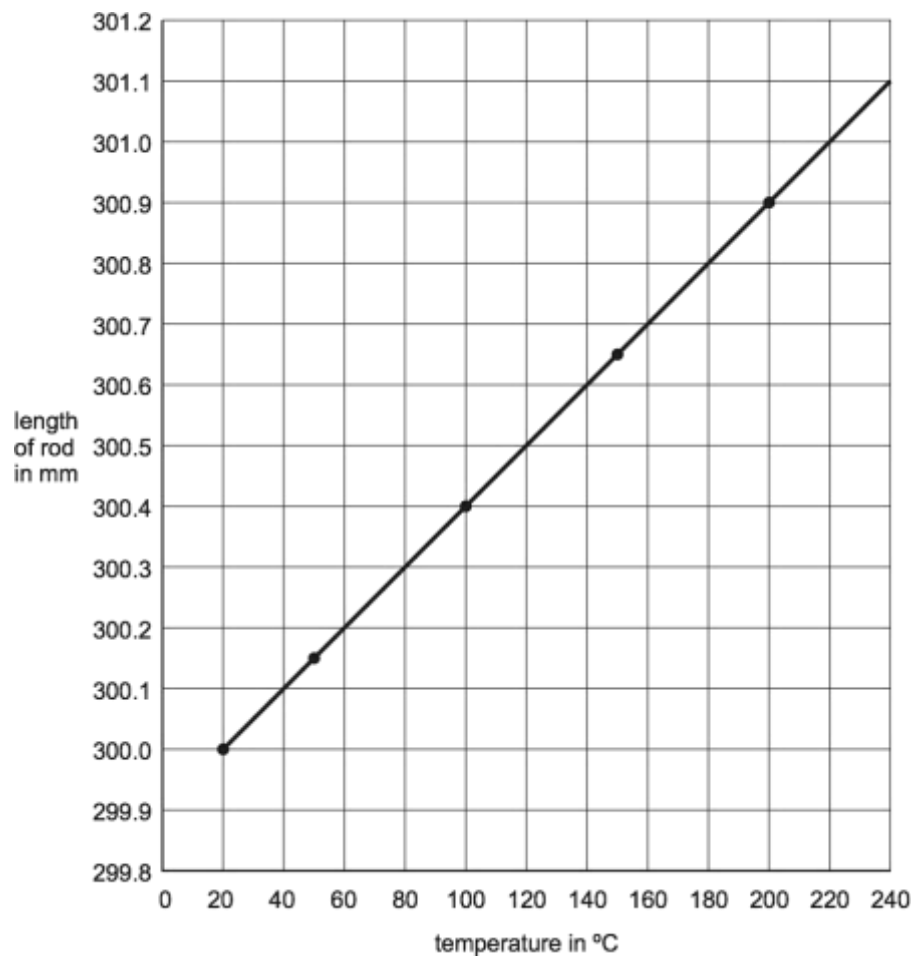
- i. What happens to the motion of the copper atoms in the rod as it is heated?

(1 mark)

- ii. What happens to the average distance between the atoms of copper when the rod is heated?

(1 mark)

- b. The graph shows the length of a copper rod at different temperatures. The rod was 300.0 mm long at room temperature.





- i. At what temperature has the rod increased its length by 1.0 mm?

_____ °C

(1 mark)

- ii. The rod was 300.0 mm long at room temperature. What will its length become if it is placed in melting ice at 0°C?

_____ mm

(1 mark)

(Total 4 marks)

8.

This question is about energy transfer processes

- a. Draw a line from each **method of energy transfer** to its **example**

method of energy transfer

example

radiation

The way the air above a heater becomes warm.

The way energy from the Sun reaches the Earth.

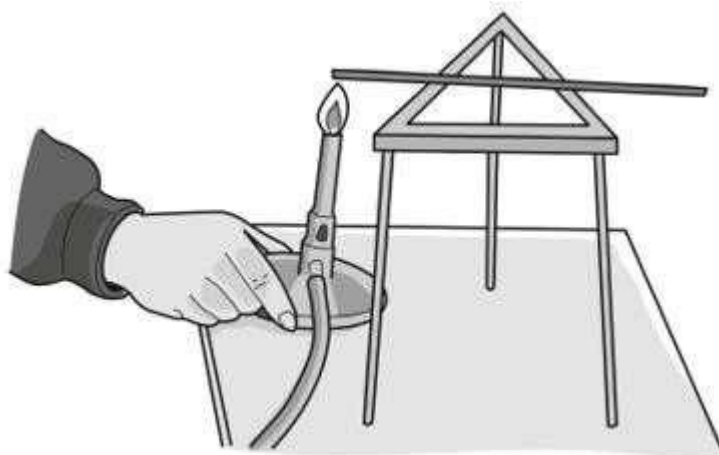
convection

The way the metal handle of a cooking pot gets hot.

(1 mark)



b. The diagram below shows a copper rod being heated



c.

Explain how heat is transferred through the copper rod

Use ideas about particles in your answer

(3 marks)

(Total 4 marks)

TOTAL FOR ASSIGNMENT 47 MARKS